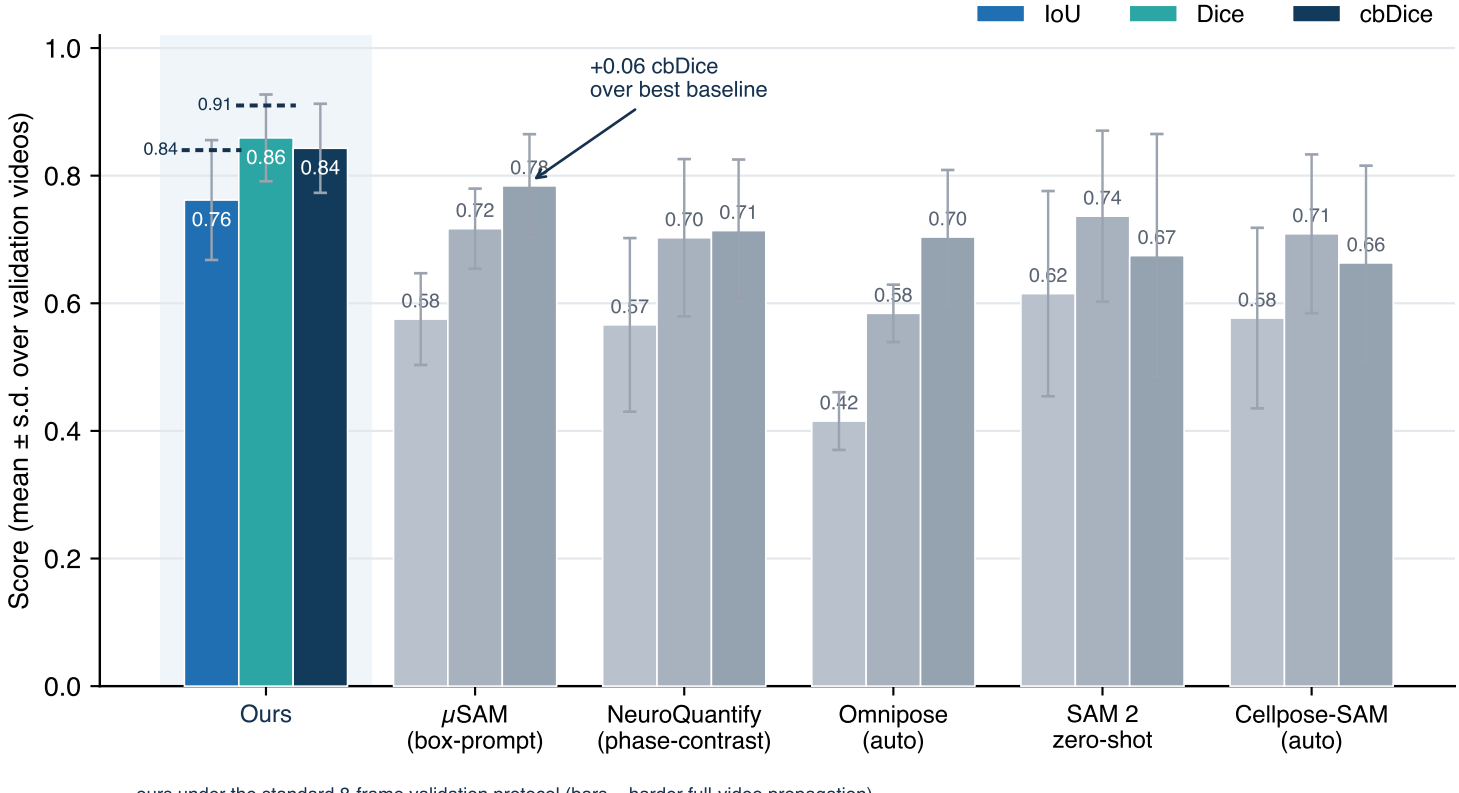
Neuron segmentation on the validation set — ours leads every metric, from a single frame-0 annotation
IoU
Dice
cbDice
Score (mean ± s.d. over validation videos)
1.0
0.8
0.6
0.4
0.2
0.0
+0.06 cbDice
over best baseline
0.91
0.84
0.86
0.84
0.76
0.58
0.72
0.78
0.57
0.70
0.71
0.42
0.58
0.70
0.62
0.74
0.67
0.58
0.71
0.66
Ours
μSAM
(box-prompt)
NeuroQuantify
(phase-contrast)
Omnipose
(auto)
SAM 2
zero-shot
Cellpose-SAM
(auto)
ours under the standard 8-frame validation protocol (bars = harder full-video propagation)